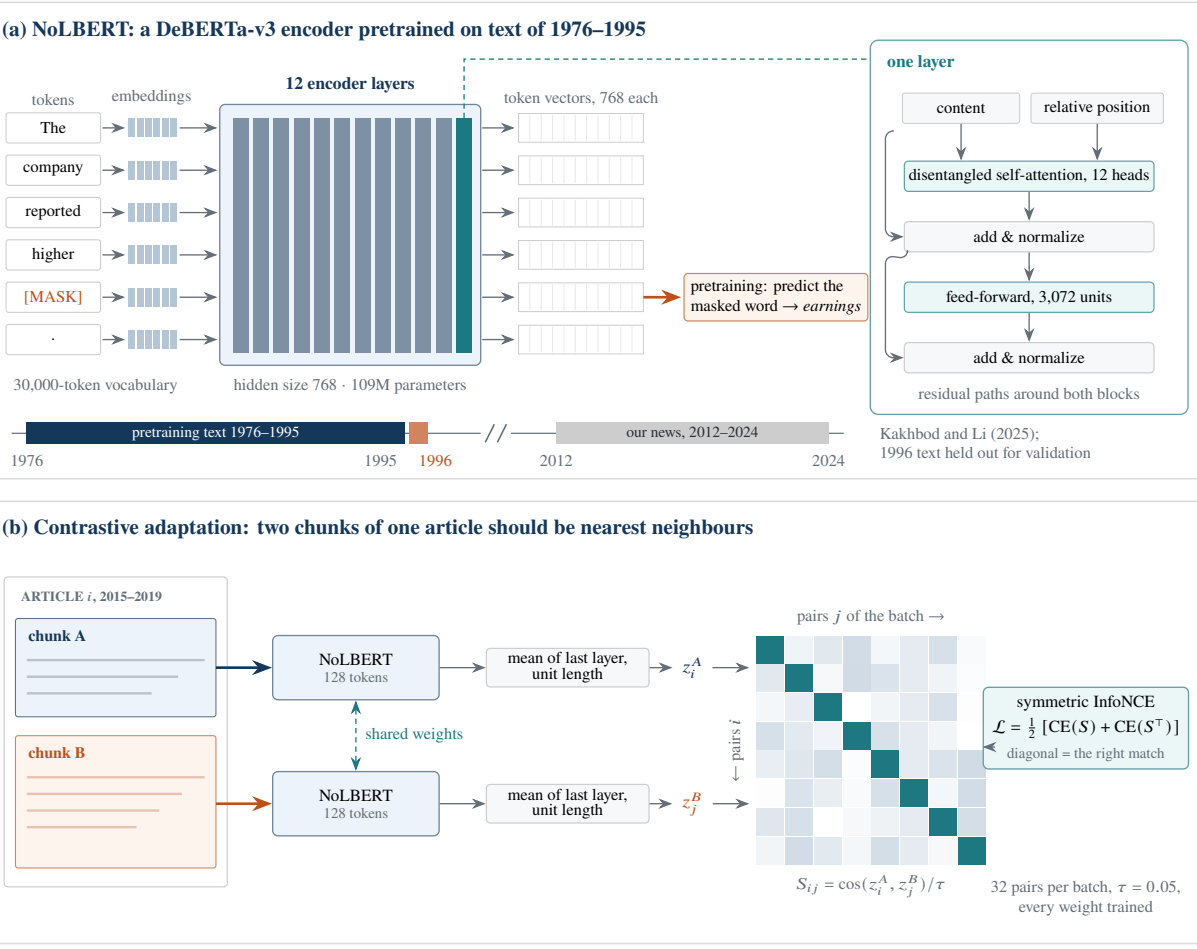

(a) NoLBERT: a DeBERTa-v3 encoder pretrained on text of 1976–1995
tokens
embeddings
12 encoder layers
token vectors, 768 each
The
company
reported
higher
[MASK]
.
pretraining: predict the masked word → earnings
30,000-token vocabulary
hidden size 768 · 109M parameters
one layer
content
relative position
disentangled self-attention, 12 heads
add & normalize
feed-forward, 3,072 units
add & normalize
residual paths around both blocks
pretraining text 1976–1995
our news, 2012–2024
1976
1995
1996
2012
2024
Kakhbod and Li (2025);
1996 text held out for validation
(b) Contrastive adaptation: two chunks of one article should be nearest neighbours
ARTICLE i, 2015–2019
chunk A
chunk B
NoLBERT
128 tokens
shared weights
NoLBERT
128 tokens
mean of last layer, unit length
mean of last layer, unit length
pairs j of the batch →
← pairs i
symmetric InfoNCE
diagonal = the right match
32 pairs per batch, τ = 0.05, every weight trained